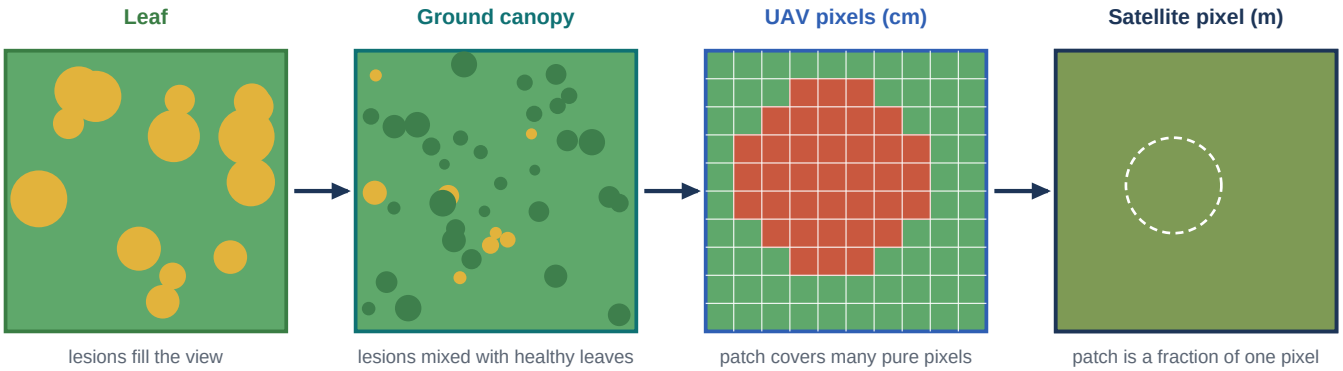
Leaf to Satellite: Crop Disease Detection Methodology

Concepts and an integrated, step-wise methodology for detecting crop disease at leaf, ground canopy, UAV and satellite level, and for linking the levels into one chain

Scope. This document covers **concepts and methodology only**. The methods of 12 published studies are merged into one workflow; each flow-chart box and table row names the studies that the step is based on (author, year). Full references are in Section 9.

1. Core concepts

- **Disease has a spectral signature.** Infection lowers chlorophyll (visible reflectance rises), damages leaf and canopy structure (NIR reflectance falls) and restricts water flow (canopy temperature rises). Each effect is read in a different spectral region (Liu Y. 2026; Poblete 2023; Yuan 2025).
- **The signal is measured at four levels.** Leaf (mm–cm), ground canopy (about 1 m²), UAV (1–10 cm pixels over fields) and satellite (0.5–20 m pixels over regions). Finer levels see the disease clearly but cover little ground; coarser levels cover large areas but mix healthy and diseased plants (Zhao 2023; Blasch 2023).
- **Upscaling means carrying knowledge up the levels.** Sensitive bands found on leaves are tested on canopies; canopy findings are converted into satellite bands; UAV maps provide pixel-sized labels for satellite models (Zheng 2018; Revill 2020; Jiang 2023).
- **Three links must match between levels:** the **bands** (spectral link), the **footprint** (spatial link) and the **calibration or model** (model link), with reflectance values made comparable (radiometric link) (Revill 2020; Yuan 2025; Poblete 2023).
- **Ground truth flows through every level.** Disease is scored in georeferenced 1 m² frames at the time of each measurement, so every level is calibrated and validated against the same plots (Zheng 2018; Guo 2021; Yuan 2025).



The same infection occupies a shrinking share of each measurement as the footprint grows, so the spectral contrast is diluted (Yuan 2025; Zhao 2023).

Figure 1. Concept of signal dilution with scale.

2. The four-level framework and its links

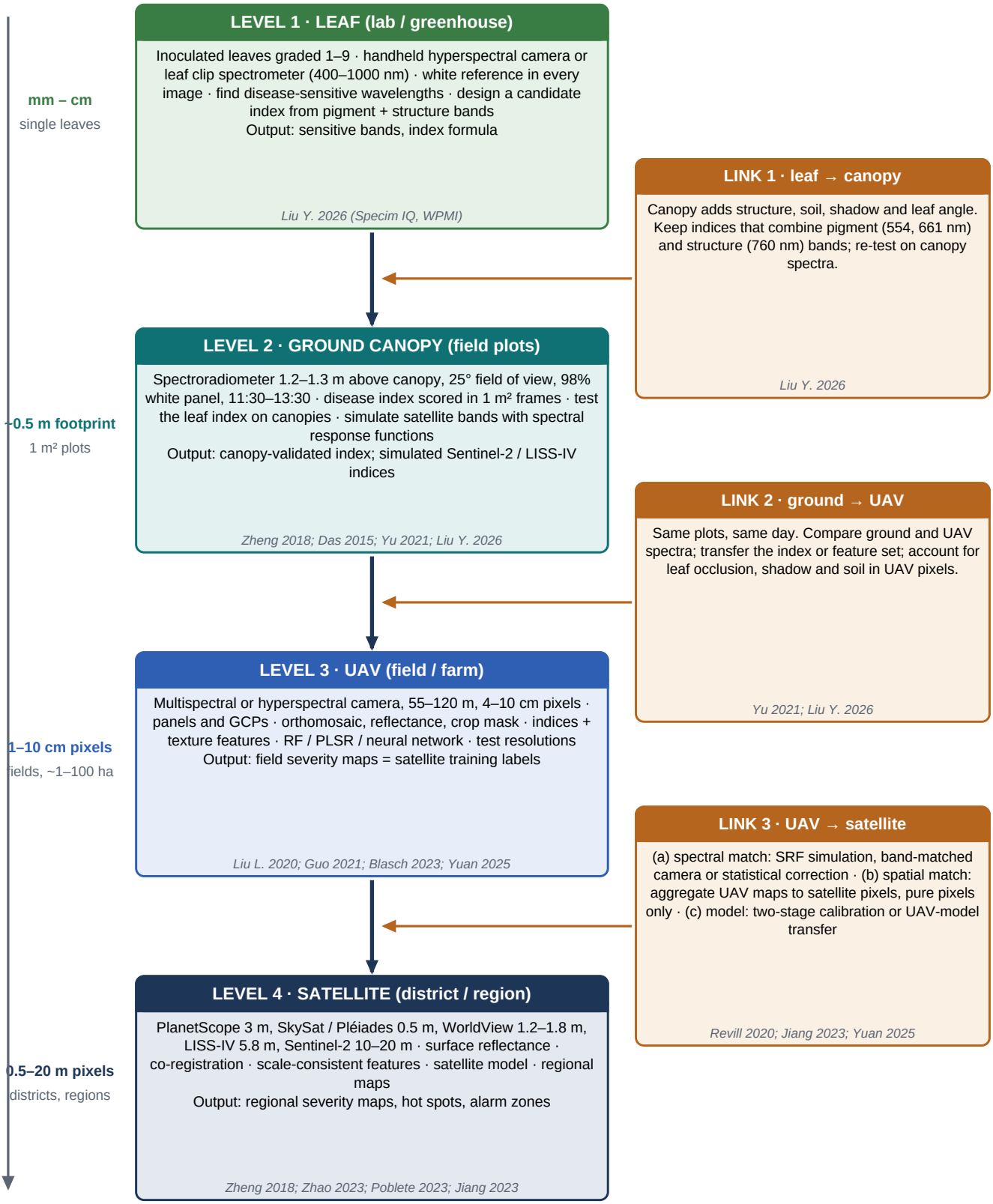


Figure 2. The four detection levels (left), the scale each works at (far left) and the method that links each level to the next (right). Arrows point in the upscaling direction.

3. Integrated step-wise methodology

All levels combined into one sequence of 16 steps. Colour shows the level of each step; amber steps are the links between levels.



Figure 3. Integrated workflow from leaf to satellite (S1–S16), with the studies each step is based on.

Step	What is done	How (method)	Based on
S1	Define symptoms	List the physiological effects of the pathogen and the spectral region each affects (visible, red edge, NIR, thermal)	Liu Y. 2026; Poblete 2023
S2	Design experiment	Inoculated, naturally infected and fungicide-protected plots; varieties of differing resistance; plot size larger than the coarsest pixel; 1 m² sample frames	Blasch 2023; Guo 2021; Liu Y. 2026
S3	Leaf spectra	Grade leaves 1–9 after inoculation; hyperspectral leaf images (400–1000 nm) with a white reference in each frame	Liu Y. 2026
S4	Sensitive bands and index	Single-band classification and band-combination search; build an index from pigment (green peak, red trough) and structure (NIR) bands	Liu Y. 2026; Zheng 2018
S5	Canopy spectra and scoring	Spectroradiometer at 1.2–1.3 m, 25° field of view, white panel, near solar noon; repeated readings averaged; disease index from 0–9 plant scores	Zheng 2018; Yu 2021; Liu Y. 2026
S6	Simulate satellite bands	Convolve canopy spectra with each sensor's spectral response function to obtain Sentinel-2, LISS-IV or other band values; re-derive indices	Zheng 2018; Das 2015

Step	What is done	How (method)	Based on
S7	UAV acquisition	Satellite-matched multispectral or hyperspectral camera; 55–120 m altitude; 75–85% overlap; solar noon ± 2 h; reflectance panels; GCPs or RTK	Revill 2020; Blasch 2023; Yuan 2025
S8	UAV processing	Structure-from-motion orthomosaic; radiometric calibration; crop/soil/shadow mask; inner plot buffer or tree-crown segmentation	Blasch 2023; Poblete 2023; Jiang 2023
S9	UAV features and model	Bands, indices and texture; feature selection; random forest, PLSR, SVM, KNN or neural network; resample to test pixel sizes	Liu L. 2020; Guo 2021; Zhao 2023
S10	Satellite acquisition	Cloud-free scene within a few days of the campaign; surface reflectance; pan-sharpening for VHR data	Zheng 2018; Blasch 2023; Poblete 2023
S11	Radiometric match	Co-register to GCPs or the UAV image; empirical-line cross-calibration or rescaling of satellite bands to UAV statistics	Yuan 2025; Poblete 2023; Zhao 2023
S12	Spatial match	Aggregate UAV maps to the satellite grid; use only pure in-field pixels away from edges and roads	Jiang 2023; Revill 2020
S13	Satellite model	Choose one: index thresholds, two-stage calibration (ground → UAV → satellite), or transfer of the UAV model using scale-consistent features	Zheng 2018; Revill 2020; Yuan 2025
S14	Map and track	Severity maps for each date; spatial hot-spot analysis to follow spread and recovery	Liu Y. 2026; Zheng 2018
S15	Validate	Independent plots or a second district; state the detection limit by growth stage and patch size	Jiang 2023; Zheng 2018; Zhao 2023
S16	Early warning	Where early stages matter, add narrow-band (PRI, SIF) and thermal (crop water stress) measurements from UAV or aircraft	Poblete 2023

4. Methodology at each level

4.1 Leaf level: finding the disease signal

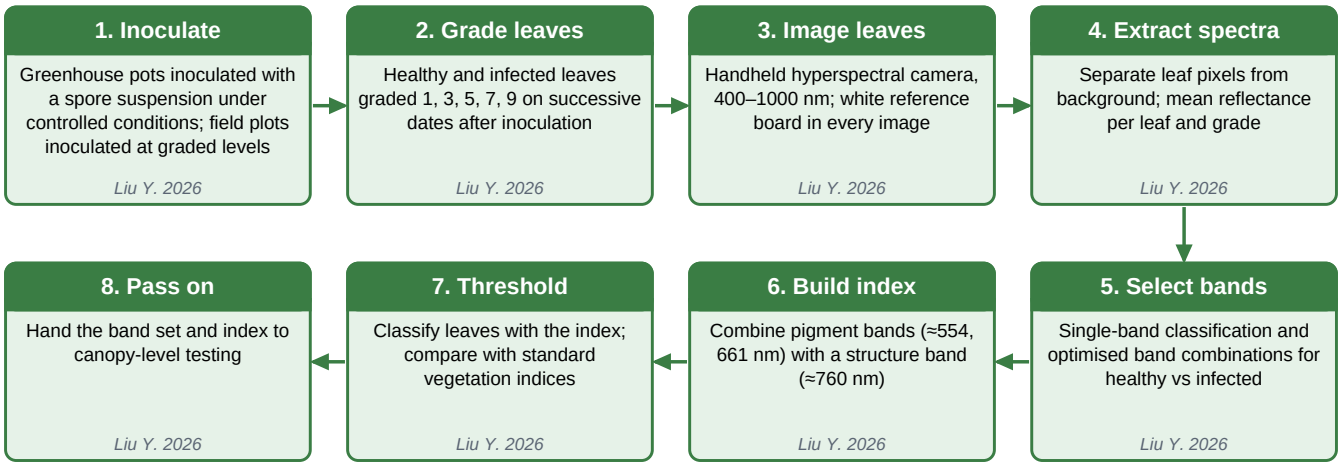


Figure 4. Leaf-level workflow.

4.2 Ground canopy level: confirming the signal in the field

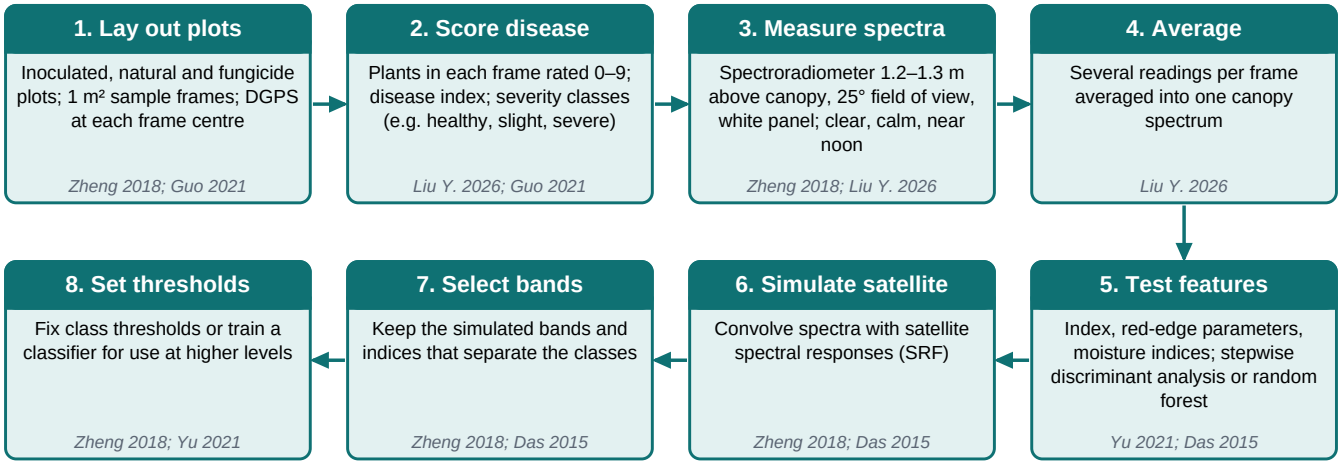


Figure 5. Ground canopy workflow.

4.3 UAV level: mapping disease across fields

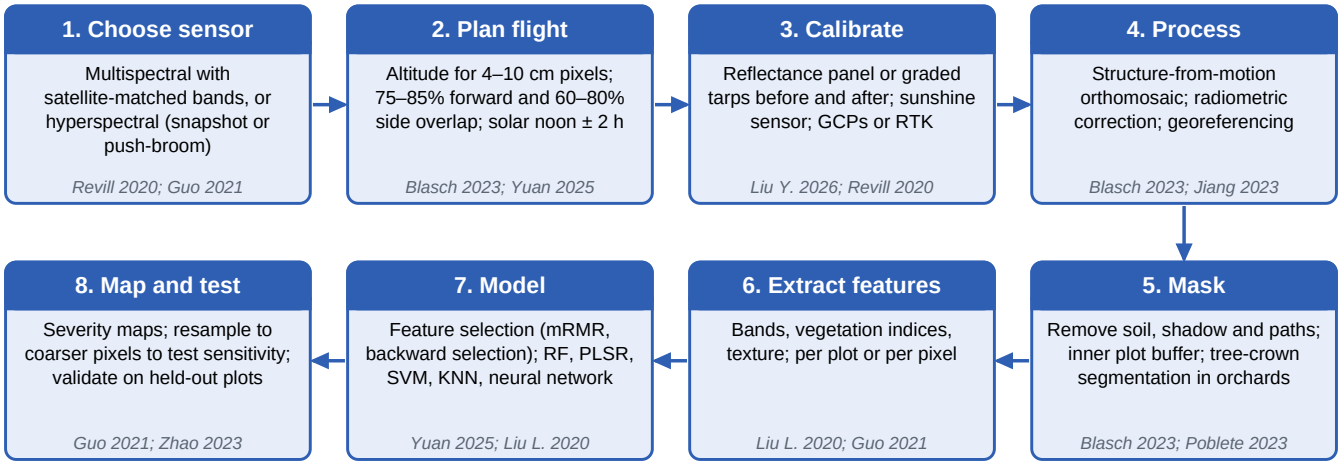


Figure 6. UAV workflow.

Study	Platform and camera	Flight / pixel	Calibration
Liu L. 2020	UAV + Cubert S185 snapshot hyperspectral	60 m, 4 cm; 80/65% overlap	Reflectance panel
Guo 2021	UAV + UHD 185 snapshot hyperspectral	1.2 cm, resampled to 3–20 cm	Reflectance panel
Yu 2021; Liu Y. 2026	DJI M600 + Resonon Pika L push-broom	120 m, ≈10 cm	White board; tarps 10/30/70%
Blasch 2023	Parrot Sequoia (G, R, red edge, NIR)	60 m, 5.6 cm; 80/80% overlap	Panel + sunshine sensor; 6 GCPs
Yuan 2025	DJI M600 + MicaSense RedEdge-M	55 m, 4 cm; 80/70% overlap	White panel
Revill 2020	DJI M600 + MAIA (Sentinel-2-matched bands)	100 m, 4.7 cm	Lambertian targets measured with ASD
Jiang 2023	eBee fixed-wing + Parrot Sequoia	100 m, 10 cm; 75% overlap	Pix4D calibration; GCPs
Zhao 2023	Hexacopter + twin Nikon D810 (RGB, NIR)	120 m, 2.9 cm; 85/60% overlap	Panels; 8 GCPs

4.4 Satellite level: regional mapping

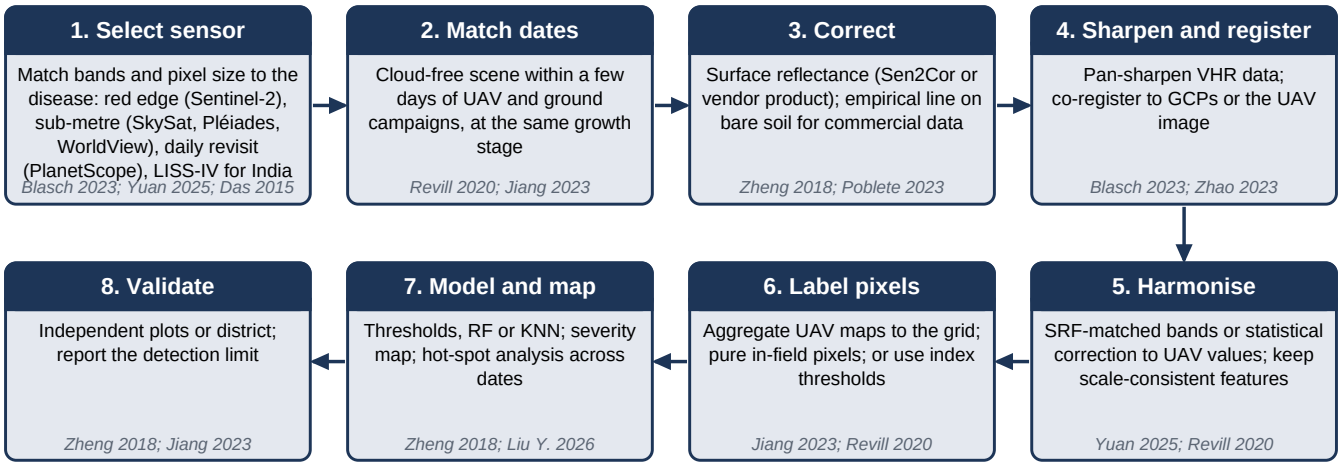


Figure 7. Satellite workflow.

5. Linking the levels

5.1 Spectral link: simulating satellite bands from spectra

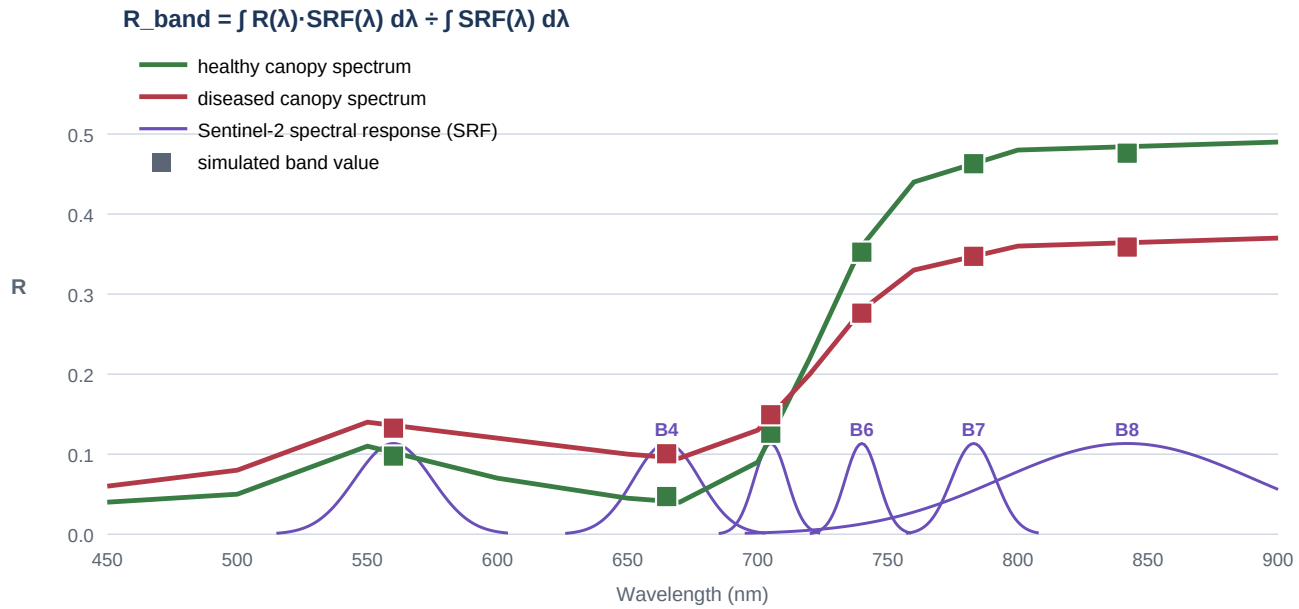
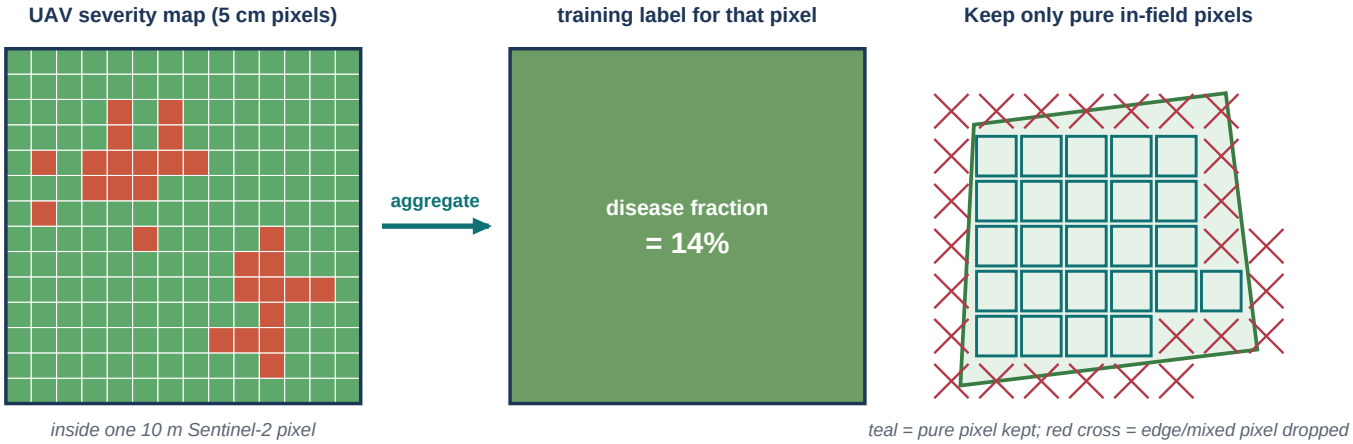


Figure 8. A canopy spectrum (line) is weighted by each band's spectral response (purple) to give the value that band would record (squares). Method of Zheng 2018 (Sentinel-2) and Das 2015 (LISS-IV). Spectra are illustrative.

- **Alternative 1:** a UAV camera whose bands copy the satellite's bands, which removes band-position differences at source (Revill 2020).
- **Alternative 2:** restrict analysis to bands every platform shares, e.g. green, red and NIR (Blasch 2023).

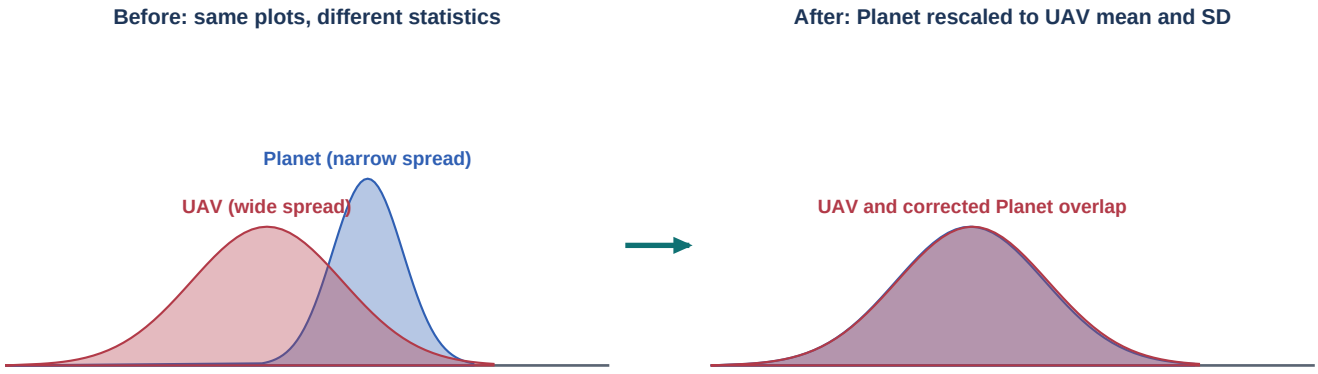
5.2 Spatial link: from UAV pixels to satellite pixels



Pixel aggregation and pure-pixel sampling (Jiang 2023; Revill 2020); 1 m plot buffers (Blasch 2023).

Figure 9. The UAV map inside one satellite pixel becomes one label (left); pixels crossing field edges or roads are dropped (right). Aggregation does not fully reproduce a real coarser sensor, whose optics also blur the image (Zhao 2023). Values are illustrative.

5.3 Radiometric link: making reflectance comparable



$$R' = (R_Planet - mean_Planet) \times (SD_UAV \div SD_Planet) + mean_UAV$$

Statistical correction (Yuan 2025). Physical alternative: empirical-line calibration on bare-soil targets against Sentinel-2 (Poblete 2023).

Figure 10. Satellite reflectance over the same plots has a narrower spread than UAV reflectance because each pixel averages healthy and diseased plants; rescaling aligns the two distributions so UAV-derived features can be applied (Yuan 2025). Schematic.

5.4 Model link: three strategies

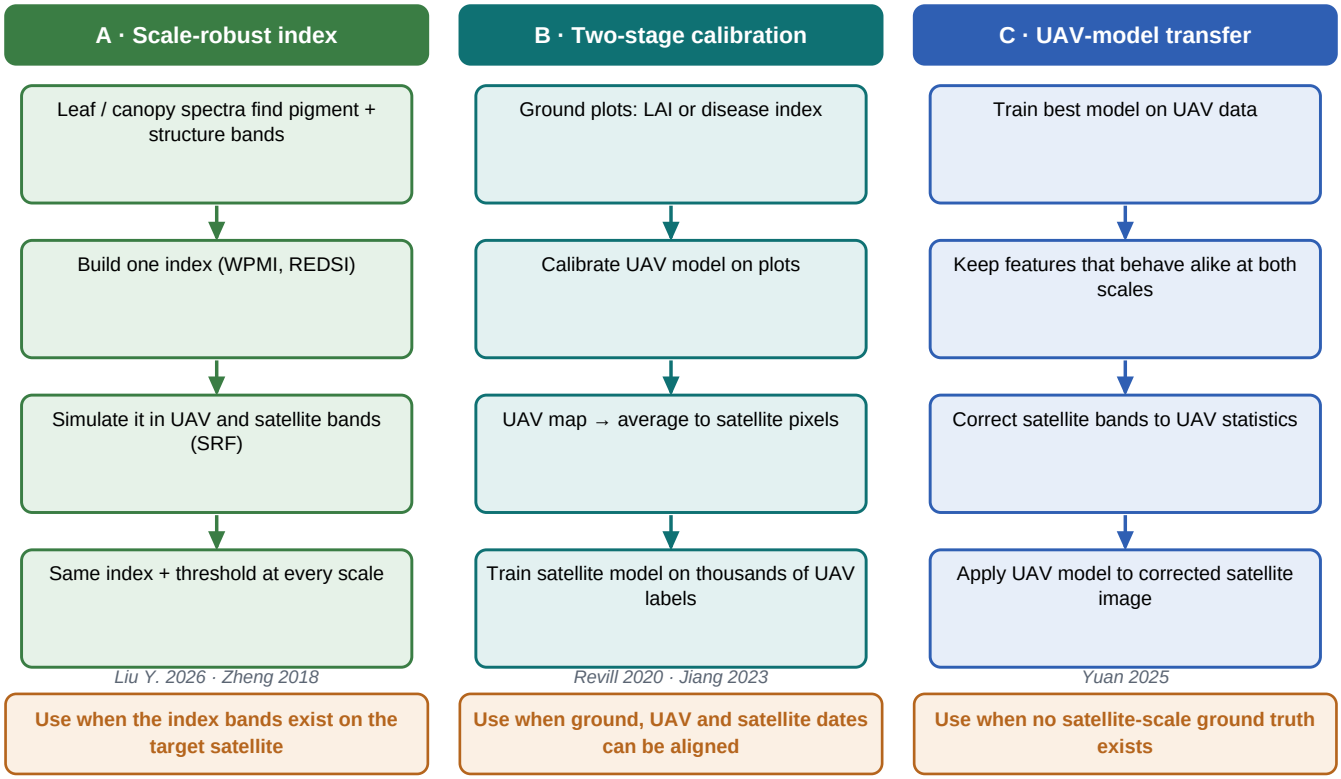
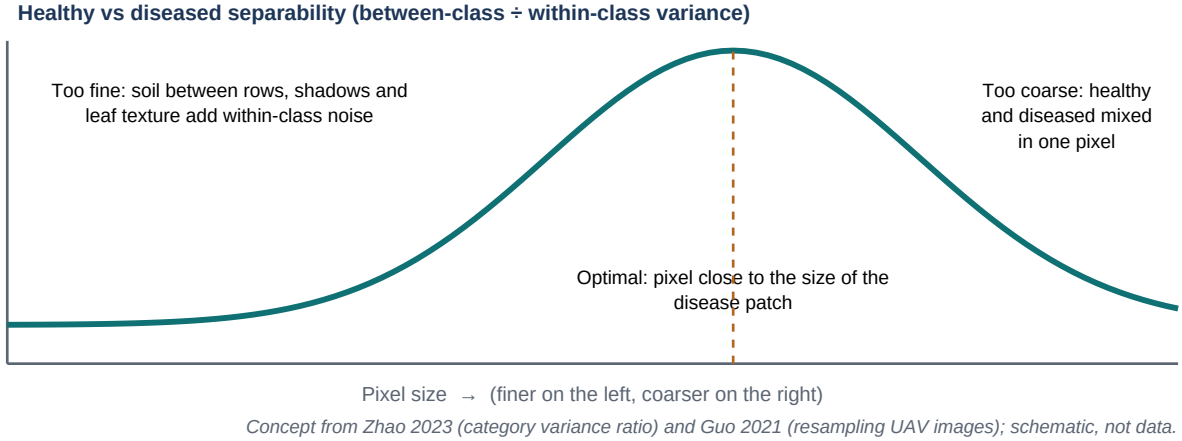


Figure 11. A: one index valid at all scales. B: the UAV converts a few ground points into many pixel-sized labels. C: a model trained on UAV data is applied to satellite data after spectral correction.

Strategy	Inputs needed	Concept	Caution
A · Scale-robust index	Leaf and canopy spectra; satellite SRFs	Physically based index combining pigment and structure bands, applied with the same logic at every level	Thresholds can shift with region, variety and stage
B · Two-stage calibration	Ground plots + UAV flights near satellite overpass	Ground → UAV model → UAV maps averaged to pixels → satellite model	Needs UAV campaigns each season
C · UAV-model transfer	UAV model; overlapping satellite scene	Align satellite bands to UAV statistics; keep features that behave alike; reuse the UAV model	Empirical correction; some detail lost at field edges

6. Scale effects to account for



Concept from Zhao 2023 (category variance ratio) and Guo 2021 (resampling UAV images); schematic, not data.

Figure 12. Concept of the optimal pixel size: separability between healthy and diseased crop rises as pixels coarsen from centimetres (less soil and row noise), peaks near the size of the disease patches, then falls as pixels mix classes.

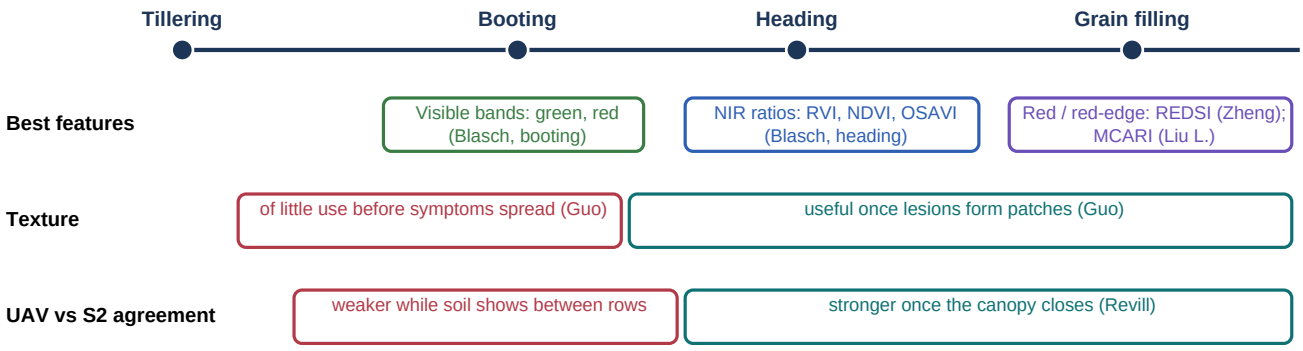


Figure 13. Useful features change with growth stage, and UAV and satellite signals agree better once the canopy closes (Blasch 2023; Guo 2021; Revill 2020; Zheng 2018; Liu L. 2020).

- **Early stages need fine spectral detail.** Broad satellite bands cannot measure traits such as PRI, SIF or anthocyanins that change first; use hyperspectral and thermal data for early detection (Poblete 2023).
- **Texture depends on resolution, spectral indices much less.** Satellite models should rely on spectral and temporal features (Guo 2021).
- **Pigment signals dominate at leaf scale, structure at canopy scale.** Indices combining both transfer best across levels (Liu Y. 2026).

7. Choosing the level for a task

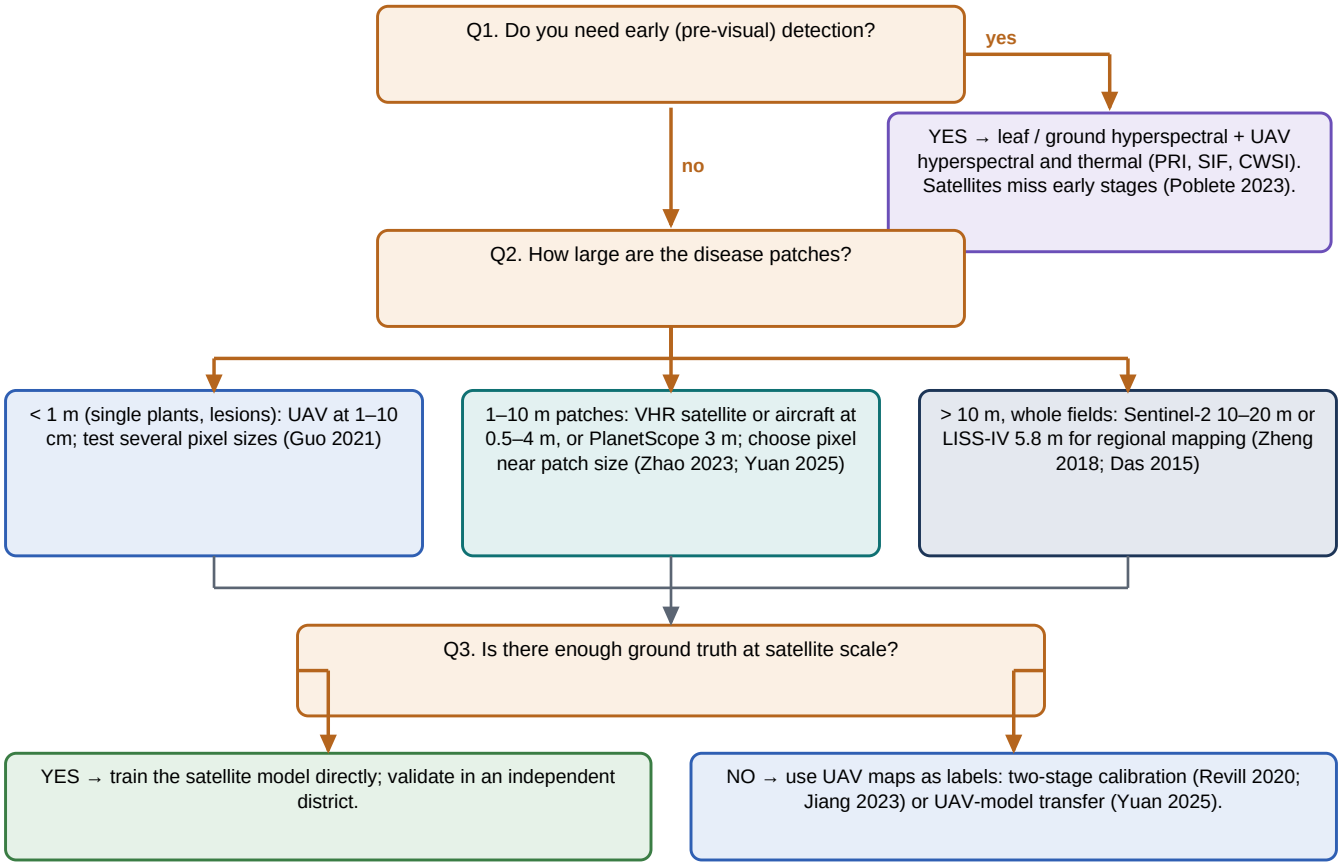


Figure 14. Decision chart linking the detection goal and patch size to the platform, and the availability of ground truth to the model-link strategy.

8. Example chain: wheat yellow rust

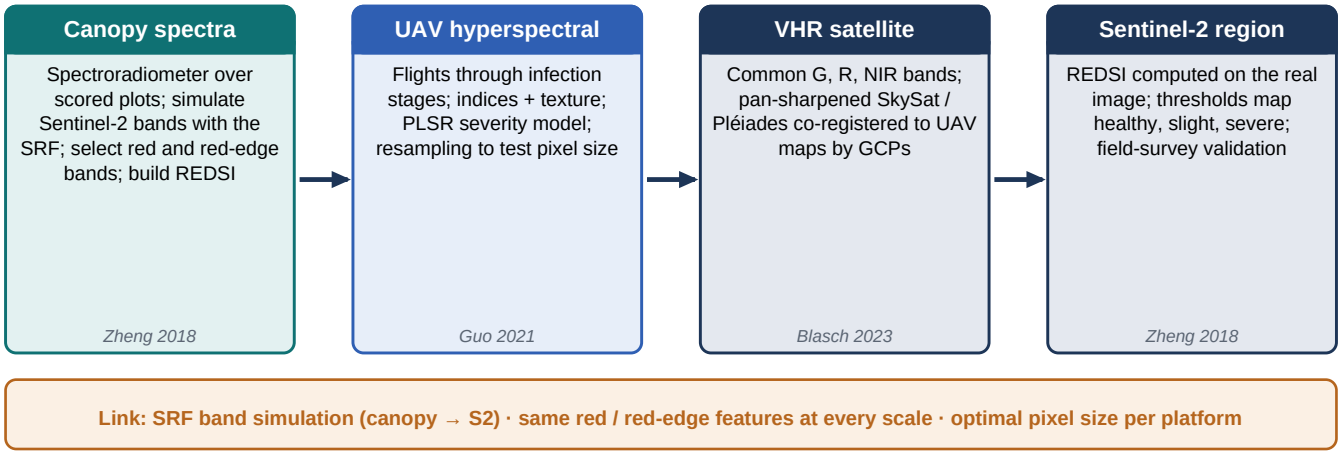


Figure 15. How the methods of three studies chain together for one disease: canopy spectra to Sentinel-2 (Zheng 2018), UAV hyperspectral monitoring (Guo 2021) and UAV-to-VHR transfer (Blasch 2023).

8.1 Key equations

Quantity	Equation	Based on
Satellite band from spectra	$R_{band} = \int R(\lambda)\cdot SRF(\lambda)\, d\lambda \div \int SRF(\lambda)\, d\lambda$	Zheng 2018; Das 2015
Red Edge Disease Stress Index	$REDSI = [(705-665)(R_{B7}-R_{B4}) - (783-665)(R_{B5}-R_{B4})] \div (2 R_{B4})$	Zheng 2018
Wheat powdery mildew index	$WPMI_G = (R760-R554)/(R661+R554)$; $WPMI_R = (R760-R661)/(R661+R554)$	Liu Y. 2026
Coefficient of infection (rust)	CI = severity (%) × host-response constant	Blasch 2023
Spectral correction	$R' = (R - mean_{sat}) \times (SD_{UAV} \div SD_{sat}) + mean_{UAV}$	Yuan 2025
Separability for pixel-size choice	CVR = between-class variance ÷ within-class variance	Zhao 2023
Pixel label from UAV map	f = diseased UAV pixels ÷ all crop UAV pixels inside the satellite pixel	Jiang 2023; Revill 2020

9. References

Blasch, G. et al. (2023). The potential of UAV and very high-resolution satellite imagery for yellow and stem rust detection and phenotyping in Ethiopia. *Scientific Reports*, 13, 16768. https://doi.org/10.1038/s41598-023-43770-y

Das, P.K., Laxman, B., Kameswara Rao, S.V.C., Seshasai, M.V.R. & Dadhwal, V.K. (2015). Monitoring of bacterial leaf blight in rice using ground-based hyperspectral and LISS IV satellite data in Kurnool, Andhra Pradesh, India. *International Journal of Pest Management*, 61(4), 359–368.

Guo, A. et al. (2021). Wheat yellow rust detection using UAV-based hyperspectral technology. *Remote Sensing*, 13(1), 123. https://doi.org/10.3390/rs13010123

Jiang, J. et al. (2023). Combining UAV and Sentinel-2 satellite multi-spectral images to diagnose crop growth and N status in winter wheat at the county scale. *Field Crops Research*, 294, 108860.

Liu, L., Dong, Y., Huang, W., Du, X. & Ma, H. (2020). Monitoring wheat Fusarium head blight using unmanned aerial vehicle hyperspectral imagery. *Remote Sensing*, 12(22), 3811. https://doi.org/10.3390/rs12223811

Liu, Y. et al. (2026). A specific vegetation index for monitoring and tracking the spatiotemporal dynamics of wheat powdery mildew from ground to unmanned aerial vehicle scales. *Journal of Remote Sensing*, 6, 0955. https://doi.org/10.34133/remotesensing.0955

Poblete, T. et al. (2023). Detection of symptoms induced by vascular plant pathogens in tree crops using high-resolution satellite data: Modelling and assessment with airborne hyperspectral imagery. *Remote Sensing of Environment*, 295, 113698. https://doi.org/10.1016/j.rse.2023.113698

Revill, A. et al. (2020). Quantifying uncertainty and bridging the scaling gap in the retrieval of leaf area index by coupling Sentinel-2 and UAV observations. *Remote Sensing*, 12(11), 1843. https://doi.org/10.3390/rs12111843

Yu, R., Ren, L. & Luo, Y. (2021). Early detection of pine wilt disease in Pinus tabuliformis in North China using a field portable spectrometer and UAV-based hyperspectral imagery. *Forest Ecosystems*, 8, 44. https://doi.org/10.1186/s40663-021-00328-6

Yuan, L. et al. (2025). Integrating UAV and high-resolution satellite remote sensing for multi-scale rice disease monitoring. *Computers and Electronics in Agriculture*, 234, 110287. https://doi.org/10.1016/j.compag.2025.110287

Zhao, H., Yang, Y., Yang, C., Song, R. & Guo, W. (2023). Evaluation of spatial resolution on crop disease detection based on multiscale images and category variance ratio. *Computers and Electronics in Agriculture*, 207, 107743. https://doi.org/10.1016/j.compag.2023.107743

Zheng, Q., Huang, W., Cui, X., Shi, Y. & Liu, L. (2018). New spectral index for detecting wheat yellow rust using Sentinel-2 multispectral imagery. *Sensors*, 18(3), 868. https://doi.org/10.3390/s18030868

Revill 2020 and Jiang 2023 study LAI and nitrogen, not disease; their scaling methods are used here because they transfer directly. Das 2015 details are from the published abstract.

Appendix

Study notes on the 12 papers behind the methodology

One card per paper: question, design, sensors, method, results, upscaling lesson, limits. Unlike the main document, this appendix includes each study's reported results.

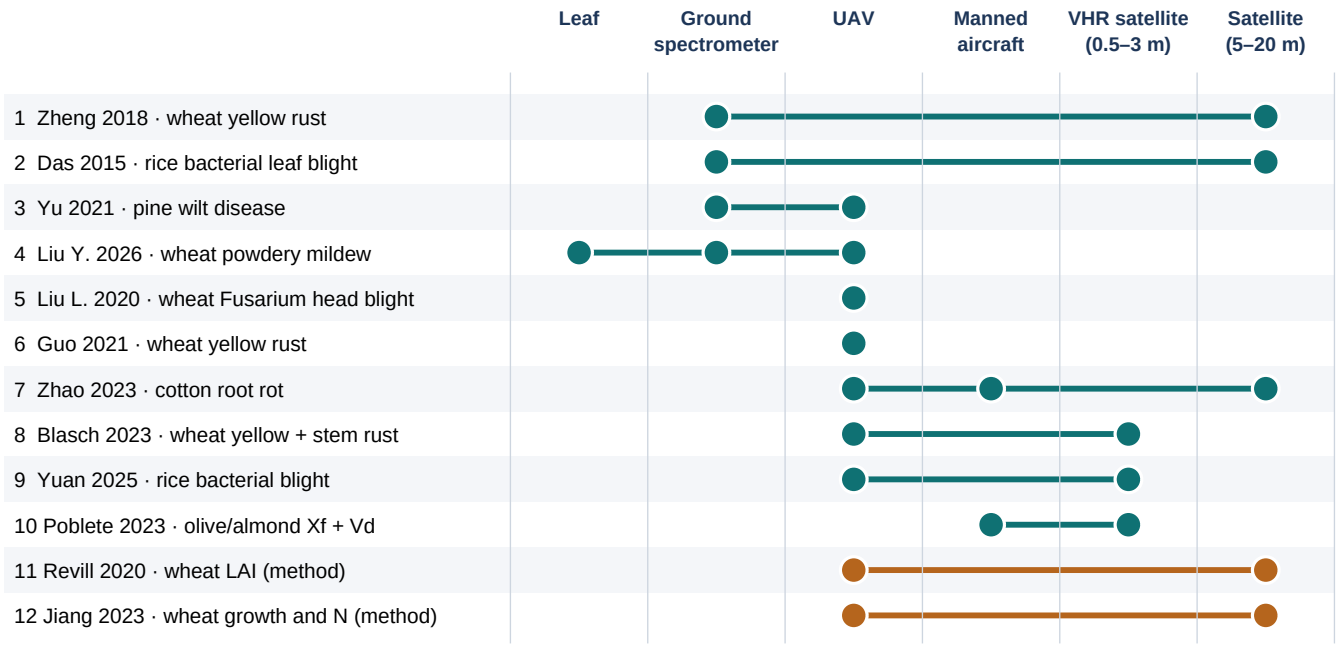
Multi-scale Crop Disease Detection: Study of 12 Papers

What each paper did, how it moved between ground, drone and satellite scales, what it found, and what the set teaches about upscaling

Bottom line. None of the 12 papers runs the full chain of ground spectrometer, UAV and satellite for one disease. The closest are Liu Y. 2026 (leaf, ground canopy and UAV for powdery mildew), Zhao 2023 (UAV, manned aircraft and Sentinel-2 for cotton root rot) and the two method papers, Revill 2020 and Jiang 2023 (ground data, UAV and Sentinel-2). Read together, they supply every link of the chain: **band simulation** (Zheng, Das), **ground-to-UAV transfer** (Yu, Liu Y.), **UAV-to-satellite transfer** (Blasch, Yuan, Zhao), **two-stage calibration** (Revill, Jiang), and the **limits** of satellites for early detection (Poblete).

1. Which scales each paper covers

Dots mark the scales at which each study collected spectral or image data; a line joins scales linked in the same study. Teal: disease studies. Amber: method papers on crop variables whose scaling approach transfers directly to disease work.



#	Paper	Crop · disease · place	What it contributes to upscaling
1	Zheng et al. 2018, *Sensors*	Wheat · yellow rust · Langfang, Hebei, China	Canopy spectra converted to Sentinel-2 bands with the sensor's spectral response; new index REDSI applied to real Sentinel-2
2	Das et al. 2015, *Int. J. Pest Management*	Rice · bacterial leaf blight · Kurnool, Andhra Pradesh, India	Indian example: ground hyperspectral bands and indices simulated for LISS-IV, then mapped at village level
3	Yu, Ren & Luo 2021, *Forest Ecosystems*	Chinese pine · pine wilt disease · Liaoning, China	Same trees measured with an ASD spectrometer and a UAV hyperspectral camera; quantifies the accuracy lost at UAV scale
4	Liu Y. et al. 2026, *J. Remote Sensing*	Wheat · powdery mildew · Beijing and Xinxiang, China	Designs one index (WPMI) on leaf spectra and shows it keeps working at ground canopy and UAV scales
5	Liu L. et al. 2020, *Remote Sensing*	Wheat · Fusarium head blight · Lujiang, Anhui, China	UAV hyperspectral severity classification (98% accuracy) at grain filling
6	Guo et al. 2021, *Remote Sensing*	Wheat · yellow rust · Langfang, Hebei, China	UAV hyperspectral monitoring by infection stage, and the effect of resampling from 1.2 cm to 20 cm

#	Paper	Crop · disease · place	What it contributes to upscaling
7	Zhao et al. 2023, *Comput. Electron. Agric.*	Cotton · root rot · Edroy, Texas, USA	UAV, manned aircraft and Sentinel-2 on the same day; finds an optimal 4 m resolution, so finer is not always better
8	Blasch et al. 2023, *Scientific Reports*	Wheat · yellow + stem rust · Kulumsa, Ethiopia	UAV (5 cm) compared with pan-sharpened SkySat and Pléiades (0.5 m) at booting and heading
9	Yuan et al. 2025, *Comput. Electron. Agric.*	Rice · bacterial blight · Fenghua, Zhejiang, China	UAV-trained model applied to PlanetScope after a spectral correction; 87.8% accuracy for 3 severity classes
10	Poblete et al. 2023, *Remote Sens. Environ.*	Olive, almond · Xylella, Verticillium · Spain, Italy, Australia	Airborne hyperspectral + thermal as benchmark for WorldView-2/3; satellites miss early stages
11	Revill et al. 2020, *Remote Sensing*	Wheat · LAI (not disease) · Scotland, UK	Two-stage calibration ground → UAV → Sentinel-2: R^2 0.88 vs 0.29 for direct calibration
12	Jiang et al. 2023, *Field Crops Research*	Wheat · biomass, N (not disease) · Xinghua, Jiangsu, China	UAV maps aggregated to Sentinel-2 pixels to train county-scale satellite models

2. Paper-by-paper study cards

Each card gives the research question, design, sensors, method, key numbers, the upscaling lesson, limitations, and a one-line summary you can say in an interview. All numbers come from the papers themselves.

1 Zheng et al. 2018 · New Spectral Index for Detecting Wheat Yellow Rust Using Sentinel-2

Ground canopy spectra → simulated Sentinel-2 bands → real Sentinel-2 image

Question	Can Sentinel-2's red-edge bands separate healthy, slight and severe yellow rust, and can an index built on simulated bands work on real imagery?
Design	• Canopy spectra: ASD FieldSpec Pro, 1.3 m above canopy, 25° field of view, grain filling 2017, Langfang (Hebei) experiment; 113 samples: 29 healthy, 39 slight (DI ≤ 20%), 45 severe (DI > 20%). • Regional test: Sentinel-2 image of 12 May 2017 (Sen2Cor, resampled to 10 m); ground survey on 9 May with 27 plots of 10 m × 10 m, five 1 m² subplots each.
Method	• Simulate Sentinel-2 bands from canopy spectra using the sensor's relative spectral response (RSR) function. • Random forest picks the sensitive bands: B4 (red), B5 (red edge 1, 705 nm), B7 (red edge 3, 783 nm). • REDSI = $[(705 - 665)(R_{B7} - R_{B4}) - (783 - 665)(R_{B5} - R_{B4})] \div (2\ R_{B4})$: the area of the red / red-edge triangle, normalised by red reflectance. • Optimal thresholds on REDSI map the three classes on the real image.
Key results	• Canopy scale: overall accuracy 84.1%, kappa 0.76; better than other common disease indices. • Regional Sentinel-2 map: overall accuracy 85.2%, kappa 0.67 against the field survey.
Upscaling lesson	Spectral response simulation is the cleanest way to carry a ground finding to a satellite: the index is designed in the satellite's own band space before any image is used.
Limits	Grain-filling stage only (late for control); one season and region; thresholds may need recalibration elsewhere.
Say it in one line	Zheng simulated Sentinel-2 bands from canopy spectra, found B4, B5 and B7 most sensitive, built REDSI, and mapped yellow rust regionally at 85% accuracy.

2 Das et al. 2015 · Bacterial Leaf Blight in Rice from Ground Hyperspectral and LISS-IV Data

Ground spectroradiometer → simulated LISS-IV → village-level maps (only the catalogue abstract was available)

Question	Can field hyperspectral signatures of bacterial leaf blight be scaled to India's LISS-IV sensor for regional monitoring?
Design	Kurnool, Andhra Pradesh. Spectroradiometer data (350–2500 nm) with chlorophyll and moisture content on healthy and blighted rice; authors from NRSC (ISRO), including V.K. Dadhwal.
Method	• Stepwise discriminant analysis finds the separating wavebands: 760, 990, 680 and 540 nm. • Twelve narrow-band indices computed; PSRI, PSSRb and red-edge position are effective. • Hyperspectral data resampled to LISS-IV resolution; PSRI and PSSRb re-computed and applied to LISS-IV imagery.
Key results	Geospatial PSRI and PSSRb maps distinguished different degrees of crop stress at village level.
Upscaling lesson	The same band-simulation logic as Zheng 2018, applied to an Indian sensor. LISS-IV has no red-edge or SWIR band, so indices must be rebuilt from green, red and NIR only.
Limits	No UAV stage; I could only read the abstract, so accuracy figures are not available here.
Say it in one line	NRSC showed ground hyperspectral indices for rice blight can be recomputed for LISS-IV and mapped at village level.

3 Yu, Ren & Luo 2021 · Early Detection of Pine Wilt Disease with a Field Spectrometer and UAV Hyperspectral Imagery

Same trees measured on the ground and from a UAV

Question	How well can each platform classify the infection stage of individual trees, and how much accuracy is lost from ground to UAV?
Design	• Chinese pine plots in Liaoning; 218 trees in four stages: healthy (76), early (54), middle (47), serious (41), defined by needle colour and resin flow. • Ground: ASD FieldSpec 4 Hi-Res NG, 350–2500 nm. • UAV: DJI M600 + Resonon Pika L, 281 bands at 2.1 nm (400–1000 nm), 120 m altitude, 18 August 2019, 12:00–12:30; 121 trees segmented from the imagery.
Method	Random forest on vegetation indices, red-edge parameters, moisture indices and their combination; stage classification and healthy vs early-stage separation.
Key results	• Ground, all features: OA 80.17%, kappa 0.73. UAV, all features: OA 74.38%, kappa 0.66. • Healthy vs early stage: red-edge parameters best on both; ground OA 80.33% (kappa 0.58), UAV OA 71.67% (kappa 0.40).
Upscaling lesson	Moving from ground to UAV costs about 6 points of overall accuracy for stage classification and more for early detection; red-edge parameters carry across scales best.
Limits	Early-stage UAV kappa (0.40) is still low; one site and date; no satellite stage.
Say it in one line	Yu found UAV hyperspectral classification of pine wilt stages slightly less accurate than ground spectra (74% vs 80%) but comparable, with red-edge features most robust.

4 Liu Y. et al. 2026 · A Specific Vegetation Index for Wheat Powdery Mildew from Ground to UAV Scales

Leaf → ground canopy → UAV canopy, three seasons

Question	Can one disease-specific index work consistently at leaf, ground canopy and UAV scales, and track spread within fields?
Design	• Greenhouse inoculation (2022) and field trials at Xinxiang (2022–2024) with susceptible Aikang 58 and mildly susceptible Kelin 618. • Leaf: Specim IQ hyperspectral camera (400–1000 nm). Ground canopy: FieldSpec HandHeld 2 at 1.2 m, 11:30–13:30, 98% panel. UAV: DJI M600 + Pika L (150 bands), 120 m, about 10 cm pixels, calibration tarps 10/30/70%. • Disease index from 20 plants per plot scored 0–9 in 1 m² frames.
Method	• Find sensitive leaf wavelengths; build $WPMI_G$ = (R760 − R554)/(R661 + R554) and $WPMI_R$ = (R760 − R661)/(R661 + R554). • Test against traditional indices at all scales; use hot-spot (Getis-Ord) analysis on UAV WPMI time series to map infection and recovery.
Key results	• Leaf classification: 85–86% (greenhouse 2022), 80–81% (field 2023, 2024). • $WPMI_G$ raised the disease-index R^2 over traditional indices by 0.06–0.92 (ground) and 0.03–0.88 (UAV) across years. • Sensitivity drops slightly from ground to UAV because of leaf occlusion, shadow and soil, and because early infection starts low in the canopy.
Upscaling lesson	An index works across scales when it combines a pigment signal (554, 661 nm) with a structure signal (760 nm): pigment dominates at leaf scale, canopy structure at canopy scale.
Limits	No satellite stage; two varieties and two sites; the authors call for validation across regions and sensors.
Say it in one line	Liu built a three-band powdery mildew index on leaf spectra that stayed better than standard indices at ground and UAV scales over three seasons.

5 Liu L. et al. 2020 · Monitoring Wheat Fusarium Head Blight Using UAV Hyperspectral Imagery

UAV hyperspectral, field scale

Design	Lujiang County, Anhui; susceptible Yangmai 25. Cubert S185 snapshot hyperspectral camera on a UAV at 60 m (4 cm pixels), 3 and 8 May 2019 (grain filling). 50 plots of 1 m²; plots under 30% infected ears labelled slight, above 30% severe.
Method	Original bands, vegetation indices and texture features; univariate threshold models; backward feature selection; back-propagation neural network improved with simulated annealing, compared with PLSR, FLDA, logistic regression, random forest and SVM.
Key results	• Red-region bands separate slight from severe infection; MCARI the best single feature (AUC 1.0). • Improved neural network: overall accuracy 98%, kappa 0.96, best of six methods.
Upscaling lesson	This is the UAV link of the Fusarium chain from the Aerospace Information Research Institute group (CAS), whose separate Sentinel-2 Fusarium index (Liu et al. 2020) completes it.
Limits	Small sample (50 plots), one cultivar, one growth stage.
Say it in one line	UAV hyperspectral imagery with an improved neural network classified Fusarium head blight severity at 98% accuracy, with MCARI the best single index.

6 Guo et al. 2021 · Wheat Yellow Rust Detection Using UAV-Based Hyperspectral Technology

UAV hyperspectral by infection stage, plus spatial resolution test

Design	Langfang Experimental Station (CAAS), Hebei; susceptible Mingxian 169. Two inoculated plots and one fungicide-protected plot (220 m² each), eight 1 m² sample regions per plot. Six UAV flights (UHD 185 snapshot camera) at 7–42 days after inoculation; 144 samples (57 healthy, 87 infected).
Method	Vegetation indices and texture features; PLSR severity models for early, mid and late infection; original 1.2 cm images resampled to 3, 5, 7, 10, 15 and 20 cm.
Key results	• Vegetation-index model best mid-infection (R^2 0.75). Texture model weak early (R^2 0.28) but strong mid and late (0.65, 0.82). • Combined model best at every stage; R^2 0.88 late. • Resolution barely affects index models but strongly affects texture models; the optimal resolution for the combined model is 10 cm.
Upscaling lesson	Spectral indices tolerate coarser pixels; texture does not. Satellites lose texture first, so satellite disease models must lean on spectral indices and time series.
Say it in one line	Guo showed UAV yellow rust models combining indices and texture work best, and that 10 cm is the optimal resolution; finer is not better.

7 Zhao et al. 2023 · Evaluation of Spatial Resolution on Crop Disease Detection (Cotton Root Rot)

UAV (2.9 cm) · manned aircraft (0.2 m) · Sentinel-2 (10 m), all on 6 July 2019

Design	25 ha cotton field near Edroy, Texas. UAV and manned aircraft carried identical twin Nikon D810 cameras (RGB + NIR via 720 nm long-pass filter); UAV at 120 m, aircraft at 1000 m. Calibration tarps (4, 16, 32, 48%) and panels; Sentinel-2 co-registered to 0.30 pixel RMS error.
Method	• Separability of healthy and root-rot areas measured with a new **category variance ratio**, CVR = between-class variance ÷ within-class variance, for NDVI, GNDVI, DVI and OSAVI. • UAV image upscaled by pixel aggregation to 0.05–10 m; CVR and image statistics tracked; ISODATA classification of original and 4 m images.
Key results	• DVI the most separating index. CVR rises from 1.5 (2.9 cm) to a peak of 60.6 at 4 m, then falls (16.3 at 5 m, 8.7 at 10 m). • Overall accuracy: original UAV 86.8% (kappa 0.69), resampled 4 m UAV 91.7%, original aircraft 96.9%, resampled 4 m aircraft 97.1% (kappa 0.92), Sentinel-2 88.3% (kappa 0.66). • Sentinel-2 caught large patches but missed small ones (root-rot producer's accuracy 59%). A UAV image upscaled to 0.2 m did not reproduce the real 0.2 m aircraft image.
Upscaling lesson	There is an optimal pixel size, roughly the size of the disease patch: too fine adds soil and row noise, too coarse mixes classes. Simple aggregation also does not reproduce a real coarser sensor, a point-spread-function effect.
Say it in one line	Zhao found 4 m the optimal resolution for cotton root rot; both centimetre UAV pixels and 10 m Sentinel-2 pixels gave lower accuracy.

8 Blasch et al. 2023 · UAV and Very High Resolution Satellite Imagery for Yellow and Stem Rust in Ethiopia

UAV 5 cm → SkySat and Pléiades pan-sharpened to 0.5 m

Design	• Kulumsa Agricultural Research Center, Ethiopia, 2020. Six varieties of differing resistance, fungicide-treated and untreated blocks, 36 plots of 5 m × 5 m; natural yellow rust and inoculated stem rust. • UAV: Parrot Bluegrass + Sequoia (G, R, red edge, NIR), 60 m, 80/80% overlap, solar noon ± 2 h, 5.6 cm (resampled to 5 cm). SkySat: 4 bands, surface reflectance, pan-sharpened to 0.5 m. Pléiades: Gram-Schmidt pan-sharpened to 0.5 m. • All datasets co-registered to six GCPs; only the common G, R and NIR bands used; 18 features; plot means with a 1 m inner buffer.
Method	Combined disease index (yellow rust + weighted stem rust), linear interpolation to a 20-day window (booting to heading), Pearson correlation and linear regression; area under the disease progress curve (AUDPC) and yield.
Key results	• Booting (day 60): green band best on all platforms, R^2adj 0.86 (UAV), 0.75 (SkySat), 0.79 (Pléiades). • Heading (day 80): NIR-based ratios take over; RVI, NDVI, OSAVI R^2adj about 0.83–0.84 (UAV) and 0.95–0.97 (SkySat, Pléiades). • Yield loss up to 78% in the susceptible variety; AUDPC vs yield r = −0.99. For AUDPC and yield, UAV features outperformed satellite features.
Upscaling lesson	Sensitive features shift with growth stage (visible bands early, NIR ratios later) and are consistent across platforms; sub-metre satellites can replicate UAV disease scoring at plot scale.
Limits	Small sample (18 data points, 6 varieties), only broad bands, one season.
Say it in one line	Blasch showed UAV results for wheat rust carry over to 0.5 m SkySat and Pléiades, with green and red bands best at booting and NDVI-type ratios at heading.

9 Yuan et al. 2025 · Integrating UAV and High-Resolution Satellite Data for Multi-Scale Rice Disease Monitoring

UAV (4 cm) model transferred to PlanetScope SuperDove (3 m)

Design	Fenghua, Ningbo, Zhejiang, 6 September 2022; six blighted fields; 90 GPS-located 1 m² points (30 healthy, 30 mild, 30 severe). DJI M600 + MicaSense RedEdge-M at 55 m (4 cm), 80/70% overlap, white panel; PlanetScope 8-band surface reflectance, upsampled 3 m → 0.3 m.
Method	• **Spectral correction**: each Planet band rescaled to the UAV distribution, $R' = (R - \text{mean}_{Planet}) \times (\text{sd}_{UAV} \div \text{sd}_{Planet}) + \text{mean}_{UAV}$. • Keep only features that respond the same way at both scales (5 bands, 16 indices; ARI and CARI dropped); rank with mRMR; train RF, SVM and KNN at each scale. • Apply the best UAV model (random forest) to corrected Planet data; compare pixel by pixel.
Key results	• Accuracy: UAV random forest 97.8%; Planet KNN 93.3%; cross-scale (UAV model on corrected Planet) 87.8%, F1 0.88. Healthy vs diseased: 98.9% for both single-scale models, 94.4% cross-scale. • Disease changes reflectance by 20–60% at UAV scale but only 5–15% at satellite scale; correction moved Planet-vs-UAV regression slopes from 0.1–0.3 to 0.8–1.3. • Errors concentrate at field edges and in mixed mild/severe areas.
Upscaling lesson	A UAV model can be reused on satellite data without satellite ground truth if the feature space is aligned and only scale-consistent features are kept, at the cost of about 10 points of accuracy.
Limits	One date, 90 samples, statistics-matching correction (not physically based).
Say it in one line	Yuan applied a UAV-trained rice blight model to spectrally corrected PlanetScope data and kept 88% accuracy without satellite-scale ground surveys.

10 Poblete et al. 2023 · Vascular Pathogen Symptoms in Tree Crops from High-Resolution Satellite Data

Airborne hyperspectral + thermal (benchmark) vs WorldView-2/3

Design	Xylella fastidiosa in olive (Apulia, Italy, 7296 trees) and almond (Majorca, Spain); Verticillium dahliae in olive (Spain, Australia); 2011–2021, severity scored 0–4 per tree.
Sensors	Airborne Headwall hyperspectral (260 bands at 6.4 nm or 371 bands at 5.8 nm) and FLIR thermal; WorldView-2/3 eight VNIR bands at 1.24–1.84 m, cross-calibrated to surface reflectance with Sentinel-2 over bare soil (empirical line, $R^2 > 0.95$).
Method	Pure sunlit crown pixels; plant traits by PRO4SAIL inversion (look-up table of 100,000 simulations, convolved with Gaussian spectral responses); SIF at 760 nm; crop water stress index from thermal; machine-learning detection at early, middle and all stages.
Key results	• WorldView detected incidence with OA 0.63–0.83, kappa 0.29–0.68. • Early stages: kappa 0.22–0.45 from satellite against 0.3–0.69 from hyperspectral data. • Adding the airborne crop water stress index to satellite data raised accuracy by 10–15% and kappa by more than 0.2. • Satellite and airborne predictions disagreed most in orchards with many early-stage trees.
Upscaling lesson	Broad satellite bands cannot measure traits such as PRI, the blue NPQI index, SIF or anthocyanins, which carry the early signal. Satellites map intermediate and advanced disease; early detection needs hyperspectral and thermal data.
Say it in one line	Poblete showed WorldView detects intermediate and advanced Xylella and Verticillium, but early symptoms need hyperspectral and thermal data.

11 Revill et al. 2020 · Bridging the Scaling Gap in LAI Retrieval by Coupling Sentinel-2 and UAV

Ground LAI → UAV (Sentinel-2-matched camera) → Sentinel-2 (20 m)

Design	Two Scottish farms, winter wheat 2017/18, three dates each, 80 ground LAI points. DJI M600 at 100 m with a MAIA camera whose nine bands match Sentinel-2 bands 1–8A (4.7 cm, resampled to 5 cm); targets measured with an ASD for absolute reflectance; Sentinel-2 L2A within ±5 days.
Method	• Compare UAV and Sentinel-2 red-edge chlorophyll index on a 20 m grid. • Gaussian process regression (bands 705 and 865 nm). Single-stage: ground LAI → Sentinel-2. Two-stage: ground LAI → UAV LAI map → averaged to 20 m → Sentinel-2 model. 40/40 train/validation split.
Key results	• UAV vs Sentinel-2 index agreement rose with growth stage: R^2 0.32 (stem elongation) to 0.75 (milk), as the canopy closed. • Single-stage: mean R^2 0.29, NRMSE 17%. Two-stage: mean R^2 0.88, NRMSE 8%; errors and biases reduced by more than 50%.
Upscaling lesson	The UAV turns a few ground points into thousands of pixel-sized labels at the right scale. Band-matched cameras remove band-position errors. The authors recommend two-stage calibration as a standard.
Say it in one line	Revill raised Sentinel-2 LAI accuracy from R^2 0.29 to 0.88 by calibrating through a band-matched UAV instead of directly from ground points.

12 Jiang et al. 2023 · Combining UAV and Sentinel-2 to Diagnose Wheat Growth and N Status at County Scale

UAV maps aggregated to Sentinel-2 grid → satellite models → county maps

Design	Seven experiments (cultivars × N rates), four farms in Xinghua County, Jiangsu, 2017–2021, jointing and booting stages. eBee fixed-wing UAV with Parrot Sequoia at 100 m (10 cm), 75% overlap; Sentinel-2 within 5 days, Sen2Cor, 20 m bands super-resolved to 10 m.
Method	• UAV models (from an earlier study) give plant dry matter and N accumulation maps; maps aggregated to 10 m by pixel aggregation; pure in-field sampling points only. • Sentinel-2 indices + weather data → RF, Lasso, ANN, PLSR with 10-fold cross-validation; independent validation on experiment 7; N nutrition index via the critical N dilution curve.
Key results	• Random forest best: dry matter R^2 0.69–0.93, N accumulation R^2 0.60–0.77 (calibration); independent validation dry matter R^2 0.42–0.65, N accumulation 0.70–0.72. • N nutrition index R^2 0.46–0.54; county map at jointing: 21% N-deficient, 43% optimal, 36% excessive.
Upscaling lesson	Pixel aggregation of UAV products is a practical way to manufacture satellite training labels; check UAV–satellite band correlations first, since some bands (UAV green, Sentinel-2 SWIR1) agree poorly.
Say it in one line	Jiang trained Sentinel-2 models on UAV maps averaged to 10 m pixels and mapped wheat N status across a whole county.

3. Cross-paper synthesis

3.1 How the papers bridge scales

Bridge	Technique	Used by	Notes
Spectral	Simulate satellite bands from hyperspectral data with the spectral response function	Zheng 2018; Das 2015; Poblete 2023 (Gaussian response for PRO4SAIL)	Design the index in satellite band space before using imagery
Spectral	Camera with bands matched to the satellite	Revill 2020	Removes band-position errors at source
Spectral	Use only bands common to all platforms	Blasch 2023 (G, R, NIR)	Simple, but loses red edge
Radiometric	Statistical correction of satellite bands to the UAV distribution (mean and standard deviation)	Yuan 2025	Fast; empirical, not physical
Radiometric	Empirical line cross-calibration using bare-soil targets and Sentinel-2	Poblete 2023	Brings commercial top-of-atmosphere data to surface reflectance
Spatial	Pixel aggregation of UAV maps to the satellite grid	Jiang 2023; Revill 2020; Zhao 2023	Aggregation does not fully reproduce a real coarser sensor (Zhao)
Spatial	Pan-sharpening satellite data to 0.5 m; plot buffers	Blasch 2023 (1 m buffer)	Keeps plot means free of path and spreader-row pixels
Spatial	Pure-pixel or crown segmentation	Jiang 2023; Poblete 2023; Yu 2021	Avoids soil, shadow and edge mixing
Model	Two-stage calibration: ground → UAV → satellite	Revill 2020; Jiang 2023	Biggest accuracy gain in the set (R^2 0.29 → 0.88)
Model	Direct transfer of a UAV model to corrected satellite data	Yuan 2025	No satellite ground truth needed; about 10 points lost
Model	Scale-robust index (pigment + structure bands)	Liu Y. 2026 (WPMI); Zheng 2018 (REDSI)	Same index, same threshold logic at several scales
Model	Physical plant traits from radiative transfer inversion	Poblete 2023	Traits transfer better than empirical indices, if the bands exist

3.2 Accuracy at each scale, side by side

Study	Finer scale	Coarser scale	What changed
Yu 2021 (pine wilt)	Ground ASD: OA 80.2%, κ 0.73	UAV hyperspectral: OA 74.4%, κ 0.66	About −6 points; early-stage κ 0.58 → 0.40
Liu Y. 2026 (powdery mildew)	Ground canopy: index R^2 up to 0.92 gain	UAV: up to 0.88 gain	Slight loss from occlusion, shadow, soil
Zheng 2018 (yellow rust)	Canopy (simulated S2): OA 84.1%, κ 0.76	Real Sentinel-2: OA 85.2%, κ 0.67	Accuracy held; kappa fell
Zhao 2023 (root rot)	UAV 2.9 cm: 86.8%; aircraft 0.2 m: 96.9%	4 m: 91.7–97.1%; Sentinel-2 10 m: 88.3%	Best at 4 m; small patches lost at 10 m
Blasch 2023 (rusts)	UAV 5 cm: R^2adj 0.83–0.86	SkySat/Pléiades 0.5 m: R^2adj 0.75–0.97	Comparable for scoring; UAV better for AUDPC and yield
Yuan 2025 (rice blight)	UAV: 97.8%	Planet: 93.3%; UAV model on Planet: 87.8%	Reflectance contrast 20–60% → 5–15%
Poblete 2023 (Xf, Vd)	Airborne hyperspectral: early κ 0.3–0.69	WorldView: early κ 0.22–0.45	Early stage lost; thermal adds 10–15%
Revill 2020 (LAI)	Ground → S2 directly: R^2 0.29	Ground → UAV → S2: R^2 0.88	Calibration route matters more than sensor

3.3 Seven lessons from the set

- **The disease signal weakens with scale.** Reflectance contrast falls from 20–60% at UAV scale to 5–15% from satellite (Yuan 2025). Indices hold up better than raw bands because ratios partly cancel the attenuation.
- **Finest is not best.** Optimal pixel sizes were 10 cm for UAV yellow rust (Guo 2021) and 4 m for cotton root rot patches (Zhao 2023). Match the pixel to the size of the disease patch.
- **Early detection does not survive to satellite scale.** Satellites detect intermediate and advanced disease; early symptoms need narrow bands (PRI, SIF, NPQI) and thermal data (Poblete 2023; Guo 2021; Yu 2021).
- **Sensitive features change with growth stage.** Visible bands at booting, NIR ratios at heading (Blasch 2023); UAV–satellite agreement improves as the canopy closes (Revill 2020).
- **Pigment plus structure transfers best.** Pigment bands dominate at leaf scale, canopy structure at canopy scale; indices combining both (WPMI, REDSI) work at several scales (Liu Y. 2026; Zheng 2018).
- **Calibrate through the UAV.** Two-stage calibration beats direct ground-to-satellite calibration (Revill 2020; Jiang 2023), and a corrected UAV model can replace satellite ground surveys (Yuan 2025).
- **Texture is lost first.** Texture features depend strongly on resolution, spectral indices hardly (Guo 2021), so satellite models should rely on spectral and temporal features.

3.4 Gaps the papers name

- No single study yet links ground spectroscopy, UAV and satellite for one crop disease; the nearest true three-level example is Hornero et al. 2020 (cited by Poblete): ground evaluation, airborne imaging spectroscopy and Sentinel-2 time series for Xylella in olive.
- Most studies cover one site, one season and few samples (Blasch: 18 data points; Liu L.: 50 plots; Yuan: 90 points).
- Multiple simultaneous diseases, and separating disease from water or nutrient stress, remain open (Blasch 2023; Poblete 2023).
- Aggregation-based upscaling ignores the sensor's point spread function, so simulated coarse images differ from real ones (Zhao 2023).

4. Putting it together: a three-scale study design for India

Example: wheat yellow rust in Punjab and Haryana (the same design suits mustard aphid in Rajasthan). Each step cites the paper it borrows from.

Step	What to do	Borrowed from
1. Plots	Inoculated and fungicide-protected plots of susceptible and resistant varieties, plots at least 5 m × 5 m so satellite pixels fit; farmers' fields for validation	Blasch 2023; Guo 2021
2. Ground spectra	ASD canopy spectra at 1.2–1.3 m, 11:30–13:30, white panel; disease index from 0–9 scores in 1 m² frames	Zheng 2018; Liu Y. 2026
3. Index design	Find sensitive bands; build a pigment + structure index; simulate Sentinel-2, LISS-IV and EOS-06 bands with their spectral responses	Liu Y. 2026; Zheng 2018; Das 2015
4. UAV flights	Multispectral camera with Sentinel-2-matched bands (or hyperspectral), solar noon ± 2 h, 80% overlap, calibration panels, GCPs; flights at tillering, booting, heading	Revill 2020; Blasch 2023
5. UAV model	Indices + texture; random forest or PLSR; test resolutions to find the optimal pixel size	Guo 2021; Liu L. 2020; Zhao 2023
6. Satellite match	Sentinel-2 (and PlanetScope for revisit) within ±5 days; surface reflectance; co-register; aggregate UAV maps to 10 m; pure in-field pixels only	Jiang 2023; Revill 2020
7. Satellite model	Two-stage calibration; keep only scale-consistent features; optionally transfer the UAV model after spectral correction	Revill 2020; Yuan 2025
8. Validate	Independent district; report OA, kappa, R^2; state the detection limit (stage and patch size)	Zhao 2023; Poblete 2023
9. Forewarn	NDVI deviation from normal + weather favourable to rust → alarm zones	Dutta et al. 2014 (from the earlier PDF)

5. Interview questions on these papers

Q. Why can't we simply train a satellite disease model on ground plots?

Ground plots are tiny points and few in number, while a satellite pixel averages 100–400 m². Revill showed direct calibration gave R^2 0.29 against 0.88 when the UAV was used as an intermediate step.

Q. How did Zheng et al. move from ground spectra to Sentinel-2?

They convolved canopy spectra with Sentinel-2's relative spectral response to simulate bands, selected B4, B5 and B7 with random forest, built REDSI, and mapped yellow rust on a real image at 85.2% accuracy.

Q. Is higher resolution always better for disease detection?

No. Zhao found 4 m optimal for cotton root rot and Guo found 10 cm optimal for UAV yellow rust. Very fine pixels add soil and row noise; coarse pixels mix classes.

Q. Can satellites detect disease early?

Poblete showed WorldView kappa of only 0.22–0.45 at early stages against 0.3–0.69 for hyperspectral data. Early signals need narrow bands such as PRI and SIF, plus thermal stress indices.

Q. How did Yuan et al. avoid satellite ground surveys?

They rescaled PlanetScope bands to match the UAV distribution, kept only features behaving the same at both scales, and applied the UAV random forest model to the corrected Planet data, reaching 87.8% for three severity classes.

Q. What makes an index work across scales?

Combining pigment-sensitive bands with a structure-sensitive NIR band, as in WPMI (554, 661, 760 nm): pigment changes dominate at leaf scale and canopy-structure changes at canopy scale.

Q. How would you adapt these methods to Indian satellites?

LISS-IV (5.8 m) lacks red-edge and SWIR bands, so build indices from green, red and NIR as Das et al. did, add Sentinel-2 for red-edge, and use EOS-06 or future hyperspectral missions for narrow-band indices.

6. References

Blasch, G., Anberbir, T., Negash, T., Tilahun, L., Belayineh, F.Y., Alemayehu, Y., Mamo, G., Hodson, D.P. & Rodrigues, F.A. Jr. (2023). The potential of UAV and very high-resolution satellite imagery for yellow and stem rust detection and phenotyping in Ethiopia. *Scientific Reports*, 13, 16768. https://doi.org/10.1038/s41598-023-43770-y

Das, P.K., Laxman, B., Kameswara Rao, S.V.C., Seshasai, M.V.R. & Dadhwal, V.K. (2015). Monitoring of bacterial leaf blight in rice using ground-based hyperspectral and LISS IV satellite data in Kurnool, Andhra Pradesh, India. *International Journal of Pest Management*, 61(4), 359–368.

Guo, A., Huang, W., Dong, Y., Ye, H., Ma, H., Liu, B., Wu, W., Ren, Y., Ruan, C. & Geng, Y. (2021). Wheat yellow rust detection using UAV-based hyperspectral technology. *Remote Sensing*, 13(1), 123. https://doi.org/10.3390/rs13010123

Hornero, A., Hernández-Clemente, R., North, P.R., Beck, P.S.A., Boscia, D., Navas-Cortes, J.A. & Zarco-Tejada, P.J. (2020). Monitoring the incidence of Xylella fastidiosa infection in olive orchards using ground-based evaluations, airborne imaging spectroscopy and Sentinel-2 time series through 3-D radiative transfer modelling. *Remote Sensing of Environment*, 236, 111480. (Cited in Poblete et al. 2023; not read here.)

Jiang, J., Atkinson, P.M., Chen, C., Cao, Q., Tian, Y., Zhu, Y., Liu, X. & Cao, W. (2023). Combining UAV and Sentinel-2 satellite multi-spectral images to diagnose crop growth and N status in winter wheat at the county scale. *Field Crops Research*, 294, 108860.

Liu, L., Dong, Y., Huang, W., Du, X. & Ma, H. (2020). Monitoring wheat Fusarium head blight using unmanned aerial vehicle hyperspectral imagery. *Remote Sensing*, 12(22), 3811. https://doi.org/10.3390/rs12223811

Liu, Y., An, L., Ma, X., Feng, H., Sun, H., Liu, G., Liu, M., Zhao, F., Yan, X., Ma, Y. & Li, M. (2026). A specific vegetation index for monitoring and tracking the spatiotemporal dynamics of wheat powdery mildew from ground to unmanned aerial vehicle scales. *Journal of Remote Sensing*, 6, 0955. https://doi.org/10.34133/remotesensing.0955

Poblete, T., Navas-Cortes, J.A., Hornero, A., Camino, C., Calderon, R., Hernandez-Clemente, R., Landa, B.B. & Zarco-Tejada, P.J. (2023). Detection of symptoms induced by vascular plant pathogens in tree crops using high-resolution satellite data: Modelling and assessment with airborne hyperspectral imagery. *Remote Sensing of Environment*, 295, 113698. https://doi.org/10.1016/j.rse.2023.113698

Revill, A., Florence, A., MacArthur, A., Hoad, S., Rees, R. & Williams, M. (2020). Quantifying uncertainty and bridging the scaling gap in the retrieval of leaf area index by coupling Sentinel-2 and UAV observations. *Remote Sensing*, 12(11), 1843. https://doi.org/10.3390/rs12111843

Yu, R., Ren, L. & Luo, Y. (2021). Early detection of pine wilt disease in Pinus tabuliformis in North China using a field portable spectrometer and UAV-based hyperspectral imagery. *Forest Ecosystems*, 8, 44. https://doi.org/10.1186/s40663-021-00328-6

Yuan, L., Yu, Q., Xiang, L., Zeng, F., Dong, J., Xu, O. & Zhang, J. (2025). Integrating UAV and high-resolution satellite remote sensing for multi-scale rice disease monitoring. *Computers and Electronics in Agriculture*, 234, 110287. https://doi.org/10.1016/j.compag.2025.110287

Zhao, H., Yang, Y., Yang, C., Song, R. & Guo, W. (2023). Evaluation of spatial resolution on crop disease detection based on multiscale images and category variance ratio. *Computers and Electronics in Agriculture*, 207, 107743. https://doi.org/10.1016/j.compag.2023.107743

Zheng, Q., Huang, W., Cui, X., Shi, Y. & Liu, L. (2018). New spectral index for detecting wheat yellow rust using Sentinel-2 multispectral imagery. *Sensors*, 18(3), 868. https://doi.org/10.3390/s18030868

Notes: figures are taken from the papers read in full, except Das et al. 2015, for which only the catalogue abstract was available. Revill 2020 and Jiang 2023 study crop variables, not disease; they are included because their scaling methods transfer directly.